

Jacob Berger

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EDUCATION

Oregon State University Corvallis, OR
Ph.D. & M.S. in Artificial Intelligence **Expected June 2030**
Honors B.S. in Applied Computer Science & Computational Mathematics **Sept 2022 – June 2026**

- **Accelerated Master's Program** (graduate coursework as undergrad), **Summa Cum Laude** (both degrees), **GPA: 3.9+**
- **Graduate Coursework:** Deep Learning, Parallel Programming, Numerical ODEs+PDEs, Math of Data Science, Computational Number Theory
- **Undergraduate Coursework:** Analysis of Algorithms, Machine Learning, Linear Algebra II

RESEARCH

Honors Thesis: Semantic Constraints in NLP Corvallis, OR
Researcher (Advisor: Dr. Lizhong Chen, STAR Lab) **June 2025 – Present**

- Led a first-author submission currently under review at COLM ("Parsimonious Decoding"), developing novel metrics to enforce rigorous semantic constraints in probabilistic language models to improve vocabulary legality.

Senior Capstone Project: AI for Home Automation Corvallis, OR
Project Team Member **September 2025 – June 2026**

- Integrating LLMs with Home Assistant for natural-language home automation; presented at OSU Engineering Research Expo.

EXPERIENCE

Oregon State University, STAR Lab Corvallis, OR
Graduate Research Assistant **June 2026 – Present**

- Conducting funded research on semantic constraints in probabilistic language models under Dr. Lizhong Chen.

NLP SaaS Startup Portland, OR
AI Research Engineer **March 2023 – Present**

- **Grant Success:** Co-authored and secured a Phase I SBIR grant (Federal Research Funding) for R&D.
- Architected end-to-end data pipelines for training and fine-tuning specialized large language models.
- Engineered and deployed cloud infrastructure on AWS, GCP, and Vercel to train and host services.

MU Lanes and Games Corvallis, OR
Head Mechanic **December 2022 – June 2026**

- Managed maintenance projects and diagnostics for complex electromechanical systems.

MU Student Activities and Advisory Board Corvallis, OR
Vice-Chairman **June 2024 – June 2025**

- Oversaw budget preparation and facilitated strategic board meetings for OSU's Memorial Union.

SKILLS

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- **Languages:** Python, C++, Java, Javascript, Matlab, Bash/Shell, x86 Assembly
 - **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, RL
 - **Systems & Infrastructure:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), GPU Passthrough
 - **Cloud & DevOps:** AWS (EC2/S3), Google Cloud Platform (GCP), Git, Vercel
 - **Languages (Human):** Fluent in Spanish (9 years of language immersion education)

AWARDS

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- **Putnam Math Competition Award:** Top 3 Finisher at Oregon State University.
 - **Finley Academic Excellence Scholarship:** \$20,000 scholarship recipient for academic merit at OSU.

EXTRACURRICULARS

- **Competitive Math:** Competitor in OSU Putnam and COMAP Math Modeling competitions (implemented Cellular Automata to analyze spatial dynamics and ecosystem resilience).
- **Competitive Sports:** Founded OSU's national collegiate bowling team (All-Star qualifier); served as Rose City Rowing Club team leader (2-time national qualifier).